Batch:	**Roll No.:**
Experiment / assignment / tutorial No.____	

Title: Mini Project - Phase 1: MVP Definition and Full-Stack Foundation.

AIM: To convert the already-selected institute AI case study into a manageable full-stack MVP and establish the first working React -> Spring Boot -> Database workflow.

Objectives:

- To define a focused problem statement, users and core features for the selected case study.
- To design the full-stack architecture, entities and REST API requirements.
- To establish the project repository and frontend/backend structure.
- To implement and demonstrate one end-to-end feature with persistent data storage.
- To specify the planned AI/intelligent feature and its integration point for a later phase.

*(Students have to perform the assigned task within the project team and demonstrate the same).

Resources used:

Tools and Technologies Used

- Frontend: React, JavaScript, HTML and CSS
- Backend: Java 17 or later, Spring Boot and REST APIs
- Database: MongoDB or another suitable project database
- Version Control: Git and GitHub
- Development Tools: Visual Studio Code / IntelliJ IDEA / Eclipse; browser; Postman/Swagger as required
- Project Context: the institute-identified AI use case already selected by the team

Expected OUTCOME of Phase 1:

CO3: Integrate front-end and back-end components using APIs and state management techniques.

CO4: Design, develop, and query relational and NoSQL databases for full-stack applications.

Students will be able to define a manageable MVP for the selected case study, design its full-stack architecture and data/API model, and demonstrate one working React -> Spring Boot -> Database workflow committed to the project repository.

Outputs to be attached / recorded:

1. Project title and selected institute case study/use-case ID.
2. One-sentence MVP statement, user roles and 3-5 core features.
3. Architecture diagram showing React, Spring Boot, database and planned AI/intelligent component.
4. Entity/resource design and REST endpoint table.
5. GitHub repository link and evidence of meaningful commits.
6. Screenshot of the working React interface.
7. Evidence of a successful React -> Spring Boot -> Database operation and persistent record.
8. Brief specification of the planned AI/intelligent feature and its integration point.

Books/ Journals/ Websites referred:

1. Course material and documentation for React, Spring Boot, database integration and REST APIs.
2. Institute AI use-case material corresponding to the case study selected by the team.

Pre Lab/ Prior Concepts:

React components/hooks, Spring Boot REST controllers, HTTP methods and JSON, database CRUD, Git/GitHub, JWT security and Swagger/OpenAPI from the completed laboratory experiments.

Project flow: User -> React Frontend -> Spring Boot REST API -> Database, with an AI/intelligent component integrated at a suitable point in a later phase.

Implementation Details:

Phase 1 System Design

Architecture: React frontend -> JSON/HTTP -> Spring Boot REST API -> persistent database; planned AI/intelligent component -> result/advisory/classification -> application workflow.

Students have to write stepwise details of implementation for the following tasks:

1. Write the project title, selected institute case study/use-case ID, one-sentence MVP statement, users and 3-5 core features.
2. Draw the proposed architecture showing React, Spring Boot, database and the planned AI/intelligent component.
3. Identify approximately 2-4 main entities/resources with important fields and relationships.
4. Define REST endpoints for one primary resource, including appropriate GET/POST/PUT/DELETE operations.
5. Create/organize the GitHub repository and implement one working React -> Spring Boot -> Database flow.
6. Specify the AI/intelligent feature: inputs, expected output, user value and proposed integration method.

Steps for execution:

1. Continue with the case study selected earlier; do not spend the session selecting a new topic.
2. Freeze the MVP, users and 3-5 core features; select one primary workflow for today.
3. Draw the architecture and identify the main data entities and API endpoints.
4. Create/organize the project repository with separate frontend and backend folders/projects.
5. Implement the backend model/repository/controller and connect it to the database.
6. Implement a React input/display component and call the backend using Axios or Fetch.
7. Demonstrate that data is stored/retrieved persistently and displayed in React.

8. Commit and push the working code and record the required evidence in this writeup.

Conclusion:

Students have to write in their own words.

Postlab questions:

1. What is the MVP of your selected case study, and which user problem does it solve?
2. Why did your team choose the implemented feature as the first end-to-end vertical slice?
3. Explain how data travels from the React interface to Spring Boot and the database in your implementation.
4. Which entities/resources are central to the project, and why?
5. What AI/intelligent functionality is planned, what inputs will it use, and what output will it produce?
6. Which parts of the final project remain to be implemented after Phase 1?